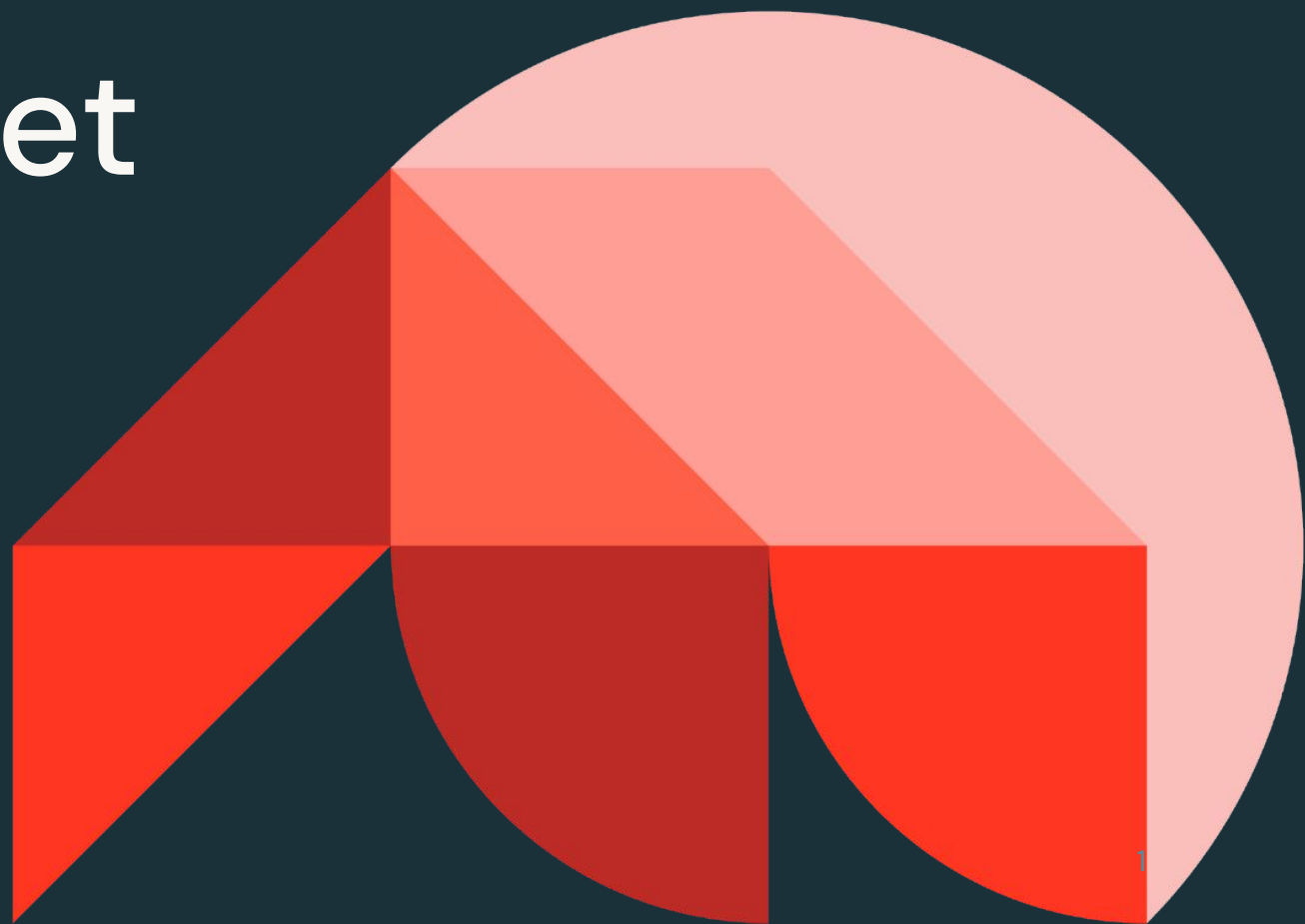




Intro to DABS – Declarative Asset Bundles



DATABRICKS ASSET BUNDLES

What?

Define Databricks resources, deployment targets, and per-target overrides, in YAML configuration.

Co-locate this with your project code for a single unit of collaboration and deployment.

How?

The **bundle** subcommand of the **Databricks CLI** lets you interact with this definition.

You can validate the configuration, deploy the configuration, and interact with resources it defines.

Where?

Commands are executed client-side, so really anywhere.

You can do development on your laptop and/or run automated production deployments from CI/CD systems such as GitHub Actions or Azure DevOps.



PROPERTIES WE'RE LOOKING FOR

- Single source of truth for code and resource configuration
- Define resources using an easy to understand format (e.g., YAML)
- Specialize resources based for target environment (dev/staging/prod)
- Ensure deployment isolation
- Easily automate deployment (CI/CD)



MAPPING TO DATABRICKS

Code

"I swear this cell was working yesterday... Did someone change it?"

- Databricks Notebooks
- Libraries (e.g. Python Wheels)
- SQL files
- JARs
- ...

Resource configuration

"The refresh job failed this morning... Was it updated to use the most recent set of expected parameters?"

- Databricks Workflows
- Delta Live Table Pipelines
- Model Serving Endpoints
- Lakehouse Monitoring
- ...

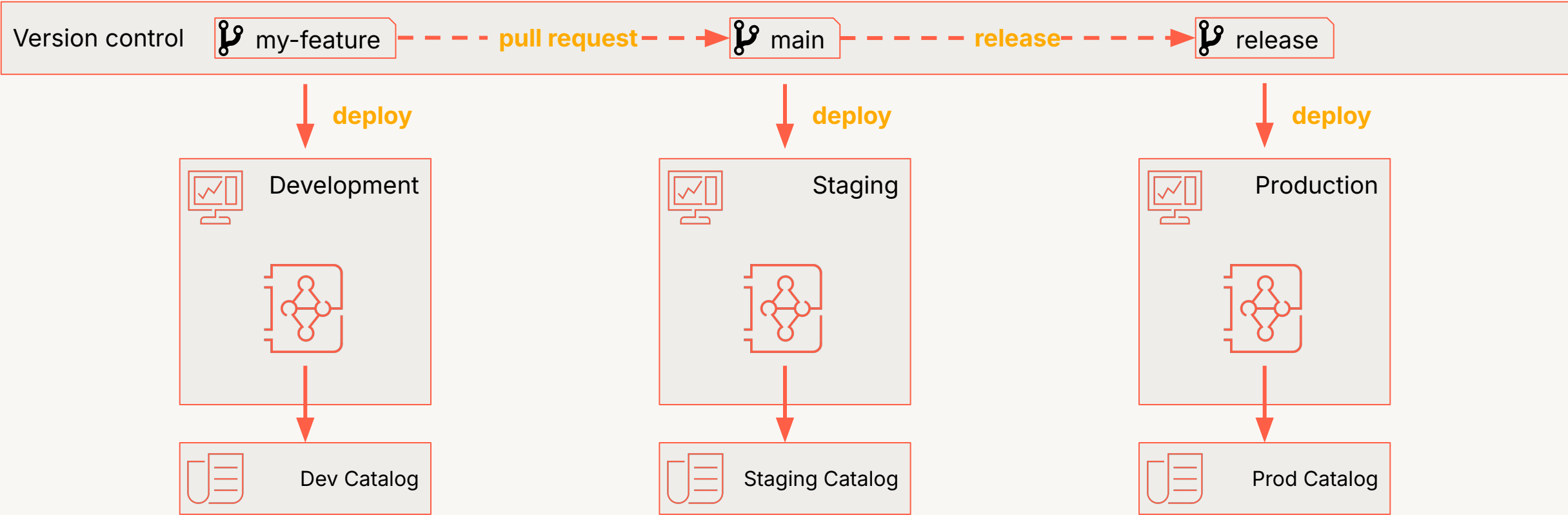
Environment separation

"We need a larger cluster in prod, who has access to update it?"

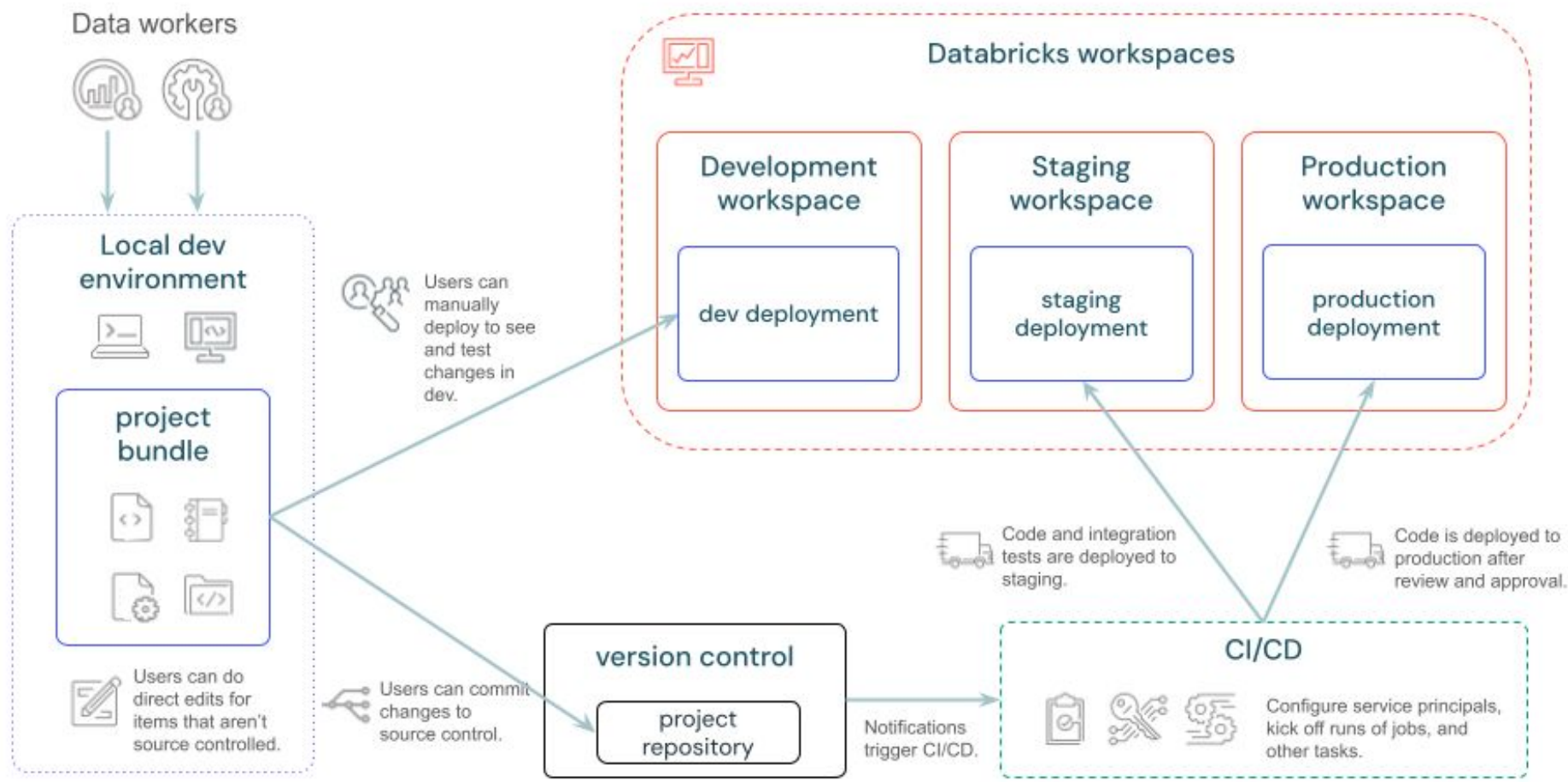
- Databricks Workspaces
- Unity Catalog (e.g. catalogs)
- Execution identity
 - User or service principal



FROM DEVELOPMENT TO PRODUCTION



Example Development Workflow



TAKING A LOOK AT `databricks.yml` (1)

- Bundle definition
- Use `include` to split configuration
- Workspace authentication
- Resource definitions (after API model)
- Local relative paths to code

```
bundle:  
  name: market_analysis  
  
include:  
  - resources/*.yml  
  
workspace:  
  host: https://my-workspace.cloud.databricks.com  
  
resources:  
  jobs:  
    market_analysis_refresh:  
      name: Run market analysis  
      tasks:  
        - task_key: run_notebook  
          job_cluster_key: cluster  
          notebook_task:  
            notebook_path: ./src/market_analysis.py
```



TAKING A LOOK AT `databricks.yml` (2)

- Define deployment `targets`
- Optional deployment modes
- Per-target overrides

```
targets:
  dev:
    mode: development
    workspace:
      host: https://dev-ws.cloud.databricks.com

  prod:
    mode: production
    workspace:
      host: https://prod-ws.cloud.databricks.com

resources:
  jobs:
    market_analysis_refresh:
      job_clusters:
        - job_cluster_key: cluster
          new_cluster:
            num_workers: 32
```



THE **bundle** CLI COMMANDS

- **Validate** to check for issues with the bundle configuration
- **Deploy** to synchronize code to the workspace and materialize the defined resources
- **Run** to interact with the deployed resources

```
$ databricks bundle validate -t dev
Warning: unknown field: num_workerzzz
  at resources.jobs.my_job.tasks[0].new_cluster
  in resources/job.yml:17:13
```

Name: market_analysis

Target: dev

Workspace:

Host: https://dev-ws.cloud.databricks.com

User: pieter.noordhuis@databricks.com

Path: ~/.bundle/market_analysis/dev

Found 1 warning

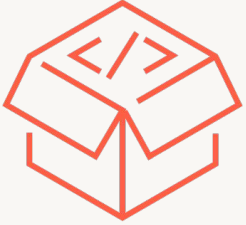


BUNDLE STATE

- Bundles use the Databricks workspace for file and state storage
- Defaults to `~/ .bundle/${bundle.name}/${bundle.target}`
 - If it **does not yet exist**; full file sync, create resources
 - If it **exists**; incremental file sync, create/update/delete resources
- Default can be modified to incorporate additional dimensions
 - E.g. include branch name for 1 deployment per feature branch



GETTING STARTED WITH BUNDLES



From scratch

- Execute `databricks bundle init`
- Select a built-in template to get started
- Leverage IDE auto-completion on the configuration schema to find what you need



From existing code and resources

- Start with an empty bundle
- Use `databricks bundle generate` to import code and resource configuration from your workspace
- Use `databricks bundle bind` to attach the bundle resource to the pre-existing resource in the workspace



DABS – Additional Resources

[Dabs Documentation](#)

[Databricks Asset Bundles: A Unifying Tool for Deployment on Databricks](#)
(~36m video)

[Substitutions and Variables in Databricks Asset Bundles](#)

[Parameterize Jobs](#)



